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EDUCATION

Mapúa University

Bachelor of Science in Computer Science, Specializing in Artificial Intelligence

Makati, Philippines

Aug. 2023 – Jul. 2026

EXPERIENCE

Artisam Labs — Centient

Lead Developer (Contract) | Next.js, TypeScript, PostgreSQL, Prisma, Redis, Stellar SDK Sep. 2026 – Present

- Lead the build of **Centient**, a human-feedback platform that won a **\$5,000** Instawards grant: contributors rank AI answers and are paid in **USDC on Stellar**, with gold tasks and agreement checks guarding quality.
- Replaced withdrawals with instant per-answer payouts after **8 of 10** balances sat stuck under the 1 USDC minimum.
- Secured payouts with **2-of-3** multisig co-signing, plus a CI race test proving **zero** double-pays.

The BLOKC (BLOCKLABS Inc.)

Software Engineering Intern

Apr. – Jun. 2026

- Led a **5-intern** engineering pod across three products, standardizing code review and test workflows to cut average feature turnaround from **4 weeks to 2 weeks** while lowering post-release error rate by **~40%**.
- Ran security-focused QA for a Learning Management System, filing **30+** reproducible issues covering IDOR, broken access control, and rate-limit bypass, cutting the pre-release defect escape rate by **~35%**.
- Built a multi-provider LLM adapter (**Groq, Anthropic, Gemini, Vertex AI, Ollama**) for Paiflow that turns prompts into Stellar payment flows: **~94%** success, **~70%** fewer malformed-output crashes.
- Shipped a **6-phase** on-chain payout system with atomic withdrawals and anti-fraud controls, stress-testing **10K+** simulated payouts on Stellar testnet with **zero** double-spends and **5/5** supervisor ratings.

PUBLICATIONS

Multi-Class Kidney Abnormality Segmentation

Aug. 2025 – Jan. 2026

AI – Deep Learning Computer Vision | IEEE ICIPCN 2026, Kathmandu University, Nepal

- Developed a YOLOv12 model to detect kidney cysts, stones, and tumors, trained on **14,761** augmented CT images.
- Achieved **mAP@0.5 of 0.946** (axial) and **0.885** (coronal), reducing radiologist workload.

RAG-Based Clinical Guideline Chatbot for Atrial Fibrillation

Dec. 2025 – Mar. 2026

AI – Natural Language Processing | IEEE CSPA 2026 | First Author

- Built a RAG chatbot (**Llama-3, Phi-3, Qwen3**) for querying **100+ pages** of ESC AF guidelines.
- Improved retrieval with MedCPT, FAISS, and BGE reranking, reaching **BERTScore F1 0.835, ROUGE-1 0.456**.
- Evaluated on **20** clinical queries, hitting faithfulness up to **8.75/10** at **~0.7s** latency and **7.9 tok/s**.

PROJECTS

Real-Time AI Exercise Coaching System

Apr. – Jun. 2026

AI – Computer Vision | Python, MediaPipe BlazePose, TFLite, ONNX, Raspberry Pi 5

- Trained a dual-head LSTM on **451,638** windows from **174 videos**: **97.15%** exercise and **92.81%** form accuracy.
- Exported to a **219 KB** TFLite model with BlazePose Lite, hitting **25–30 FPS** on a Raspberry Pi 5.
- Built a **~115 MB** on-device RAG chatbot over PDF fitness manuals, removing PyTorch from the Pi runtime.

ACTIVITIES

Pink Raft — 1st Runner-Up, Stellar Hackathon 2026

May 2026

Full-Stack Engineer & AI Integration Lead

- Built a no-code Stellar/Soroban payment flow builder that deploys live smart contracts in under **60 seconds**.
- Cut AI response latency **~80%** (10s → 2s) via schema hardening; shipped WebAuthn 2FA and non-custodial wallets.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL, R, HTML/CSS

Frameworks & Libraries: React, Next.js, React Native, Expo, Node.js, FastAPI, Flask, Stellar SDK, Soroban

AI/ML: RAG, NLP, Computer Vision, Object Detection, PyTorch, TFLite, ONNX, OpenCV, NumPy, pandas

Developer Tools: Git, Docker, Railway, Sentry, Streamlit, NVIDIA CUDA

Cloud & Databases: PostgreSQL, Prisma, Redis, Firebase, Appwrite, Google Cloud Platform, RESTful APIs